

Berka Database

Modeling & Analyzing

A relational banking data project — from schema to insight

Project Objective

This project applies mathematical modeling and statistical analysis on the Berka Database, identifying the core variables of the banking system and the relationships between them and summarizing what each variable reveals — forming a complete structural representation of the bank.

The goal was not to answer a single question, but to model the business as an integrated whole. So, the final output is **an interactive dashboard** that visualizes the bank, from which clear insights were extracted.

Data Overview

The data is from a database called **Berka**. It is a real-world relational banking dataset from a Czech bank, representing the full operational structure of the bank — customers, accounts, transactions, loans, cards, orders, and regional demographics. The data spans the period from 1993 to 1998, and it is a well-known dataset in the data science and academic research community.

The data was accessed by connecting to a public academic server hosted by Czech Technical University in Prague [relational.fel.cvut.cz] via MySQL Workbench, from which the required data was extracted and transformed using SQL [see SQL file]. The result was 7 clean, structured tables exported as CSV files [see CSV files] — ready for analysis.

The Modeling Approach

The modeling process started from the business itself. **The account was identified as the core entity** — it is the point where every relationship between the bank and its customers exists. Everything else was built around it.

For each account, the model captures: the owner's demographics (gender, age, region), financial behavior (incoming, outgoing, balance), and product relationships (loans, orders). This forms a complete profile of the bank's activity — not as isolated tables, but as a connected whole centered on the account.

Key Observations & Insights

From the dashboard, the data reveals the following:

The bank holds 4,500 accounts. Account creation across the period shows an inconsistent acquisition pattern — peaks and drops rather than steady growth.

The customer base is balanced by gender. By age, **middle aged** owners dominate at 36%, followed by **adults** at 29%.

Monthly, these accounts bring approximately **78M Kč** into the bank. Approximately 21M Kč go out as monthly transfers and 60M Kč as cash withdrawals. In return, the bank collects **only 77K Kč in service fees**.

3,758 accounts — **84%** of the base — have active orders, dominated by utility payments. The bank is deeply embedded in its customers' recurring financial lives.

682 accounts have loans — only **15%** of the total base. **89%** of those are in good standing.

Regionally, **Prague** stands clearly apart — highest average salary, lowest unemployment, highest entrepreneurship density. Yet the bank's heaviest account presence is in North Moravia, which has the highest unemployment. The bank is strongest where the economy is weakest, and underleveraged where the money is.

The bank has a stable, engaged customer base that brings in significant monthly funds. But it captures almost none of that value through products. The accounts exist. The demographics are right. What is missing is activation — more loans to the right existing customers, and a stronger presence in the regions where the real economic power is.